Number of Stories	98,161
Vocabulary size	33,215
Average number of words	50

Table 2: Statistics of the ROCStories dataset.

commonsense stories as training data, and additional 1,817 stories for development and test, respectively. The stories in the corpus are five-sentence stories that capture a rich set of causal and temporal commonsense relations between daily events, making them a good resource for training storytelling models. Table 2 shows the statistics of ROCStories dataset. Since only the training set of the ROCStories corpus contains titles, which we need as input. We split the original training data into 8:1:1 for training, validation, and testing.

Baselines

To evaluate the effectiveness of the plan-and-write framework, we compare our methods against representative baselines without a planning module.

Inc-S2S denotes the incremental sentence-to-sentence generation baseline, which creates stories by generating the first sentence from a given title, then generating the i-th sentence from the title and the previously generated i-1 sentences. This resembles the *dynamic* schema without planning. We use a Seq2Seq model with attention (Bahdanau, Cho, and Bengio 2015) to implement the Inc-S2S baseline, where the sequence to sequence model is trained to generate the next sentence based on the context.

Cond-LM denotes the conditional language model baseline, which straightforwardly generates the whole story word by word from a given title. Again we use a Seq2Seq model with attention as our implementation of the conditional language model, where the sequence to sequence model is trained to generate the whole story based on the title. It resembles our *static* schema without planning.

Hyper-parameters

As all of our baselines and the proposed methods are RNN-based conditional generation models, we conduct the same set of hyper-parameter optimization for them. We train all the models using stochastic gradient descent (SGD). For the encoder and decoder in our generation models, we tune the hyper-parameters of the embedding and hidden vector dimensions and the dropout rate by grid search. We randomly initialize the word embeddings and tune the dimensions in the range of [100, 200, 300, 500] for storyline generation and [300, 500, 1000] for story generation. We tune the hidden vector dimensions in the range of [300, 500, 1000]. The embedding and hidden vector dropout rates are all tuned from 0 to 0.5, step by 0.1. We tune all baselines and proposed models based on BLEU scores (Papineni et al. 2002) on the validation set. Details of the best hyper-parameter values for each setting are given in Appendix.

Evaluation Metrics

Objective metrics. Our goal is generating human-like stories, which can pass the Turing test. Therefore, the evaluation metrics based on n-gram overlap such as BLEU are not suitable for our task[5]. To better gauge the quality of our methods, we design novel automatic evaluation metrics to evaluate the generation results at scale. Since neural generation models are known to suffer from generating repetitive content, our automatic evaluation metrics are designed to quantify diversity across the generated stories. We design two measurements to gauge inter- and intra-story repetition. For each sentence position i, the inter-story r_e^i and intra-story r_a^i repetition rate are computed as follows:

$$
\begin{aligned}
r_e^i &= 1 - \frac{T(\sum_{j=1}^{N} s^{ji})}{T_{all}(\sum_{j=1}^{N} s^{ji})} \\
r_a^i &= \frac{1}{N} \sum_{j=1}^{N} \left[\frac{\sum_{k=1}^{i-1} T(s^i \cap s^k)}{(i-1) * T(s^i)} \right]^j
\end{aligned}
\tag{3}
$$

where $T(\cdot)$ and $T_{all}(\cdot)$ denote the number of distinct and total trigrams[6], respectively. s^{ji} stands for the i-th sentence in j-th story; $s^i \cap s^k$ is the distinct trigram intersection set between sentence s^i and s^k. Naturally, r_e^i demonstrates the repetition rate between stories at sentence position i; r_a^i embodies the average repetition of sentence s^i comparing with former sentences in a story.

We compute the aggregate scores as follows:

$$
\begin{aligned}
r_e^{agg} &= 1 - \frac{T(\sum_{j=1}^{N} \sum_{i=1}^{m} s^{ji})}{T_{all}(\sum_{j=1}^{N} \sum_{i=1}^{m} s^{ji})} \\
r_a^{agg} &= \frac{1}{m} \sum_{i=1}^{m} r_a^i
\end{aligned}
\tag{4}
$$

where $\sum_{j=1}^{N} \sum_{i=1}^{m} s^{ji}$ is the set of N stories with m sentences. In our experiments, we set $m = 5$. r_e^{agg} indicates the overall repetition of all stories.

Subjective metrics. For a creative generation task such as story generation, reliable automatic evaluation metrics to assess aspects such as interestingness, coherence are lacking. Therefore, we rely on human evaluation to assess the quality of generation. We conduct pairwise comparisons, and provide users two generated stories, asking them to choose the better one. We consider four aspects: *fidelity* (whether the story is on-topic with the given title), *coherence* (whether the story is logically consistent and coherent), *interestingness* (whether the story is interesting) and *overall user preference* (how do users like the story). All surveys were collected on Amazon Mechanical Turk (AMT).

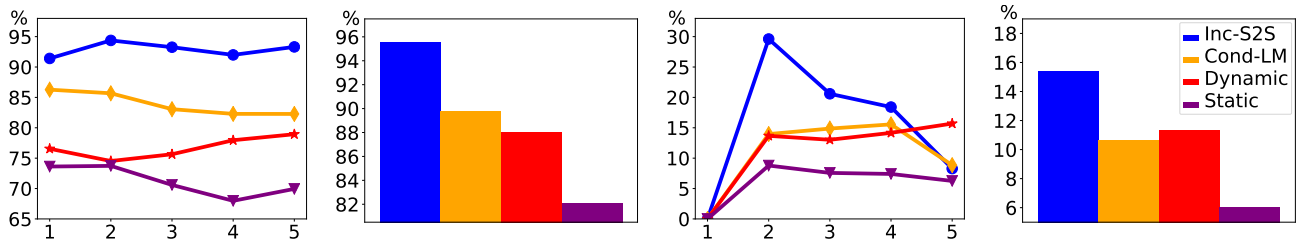
Results and Discussion

Objective evaluation

We generate 9816 stories based on the titles in the held-out test set, and compute the repetition ratio (the lower, the better) as described in Eq. 3 and Eq. 4 to evaluate

[5]Our plan-and-write methods also improve BLEU scores over the baseline methods, more details can be found in Appendix.

[6]We also conduct the same computation for four and five-grams and observed the same trends. The Spearman correlation between this measurement and human rating is 0.28.



(a) Inter-story repetition curve by sentence.

(b) Inter-story aggregate repetition scores.

(c) Intra-story repetition curve by sentence.

(d) Intra-story aggregate repetition scores.

Figure 3: Inter- and intra-story repetition rates by sentences (curves) and for the whole stories (bars), the lower the better. As reference points, the aggregate repetition rates on the human-written training data are 34% and 0.3% for the inter- and intra-story measurements respectively.

Choice %	Dynamic *vs* Inc-S2S			Static *vs* Cond-LM			Dynamic *vs* Static		
	Dyna.	Inc.	Kappa	Static	Cond.	Kappa	Dyna.	Static	Kappa
Fidelity	**35.8**	12.9	0.42	**38.5**	16.3	0.42	21.47	**38.00**	0.30
Coherence	**37.2**	28.6	0.30	**39.4**	32.3	0.35	28.27	**49.47**	0.36
Interestingness	**43.5**	26.7	0.31	**39.5**	35.7	0.42	34.40	**42.60**	0.35
Overall Popularity	**42.9**	27.0	0.34	**40.9**	34.2	0.38	30.07	**50.07**	0.38

Table 3: Human evaluation results on four aspects: fidelity, coherence, interestingness, and overall user preference. Dyna., Inc., and Cond. is the abbreviation for Dynamic schema, Inc-S2S, and Cond-LM respectively. We also calculate the Kappa coefficient to show the inter-annotator agreement.

the diversity of the generated system. As is shown in Figure 3, the proposed plan-and-write framework significantly reduces the repetition rate and generates more diverse stories. For inter-story repetition, plan-and-write methods significantly outperform all non-planning methods on individual sentences and aggregate scores. For the intra-story repetition rate, plan-and-write methods outperform their corresponding non-planning baselines on aggregate scores. However, the dynamic schema generates more repetitive final sentences than the baselines.

Subjective evaluation

For human evaluation, we randomly sample 300 titles from the test data, and present a story title and two generated stories at a time[7] to the evaluators and ask them to decide which of the two stories is better[8]. There are 233 Turkers[9] participated in the evaluation. Specifically, 69, 77, and 87 Turkers evaluate the comparison between Dynamic and Inc-S2S, Static and Cond-LM, Dynamic and Static, respectively.

Table 3 demonstrates the results of human evaluation. Similar to the automatic evaluation results, both dynamic and static schema significantly outperform their counterpart baseline in all evaluation aspects, thus demonstrating the effectiveness of the proposed plan-and-write framework. Among them, the static schema shows the best results.

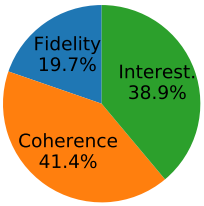


Figure 4: The regression coefficient that shows which aspect is more important in human evaluation of stories.

To understand why people prefer one story over another, we analyze how people weigh the three aspects (fidelity, coherence, and interestingness) in their preference for stories. We train a linear regression using the three aspects' scores as features to predict the overall score. We fit the regression with all human assessments we collected. The weight assigned to each aspect reflects their relative importance. As evident in Figure 4, coherence and interestingness play important roles in the human evaluation, and fidelity is less important.

Analysis

The previous sections examine the overall performance of our methods quantitatively. In this section, we qualitatively analyze our methods with a focus on comparing the dynamic and static schema.

Storyline analysis. First, we measure the quality of the generated storylines, and the correlations between a storyline and the generated story. We use BLEU scores to measure the quality of the storylines, and an embedding-based metric (Liu et al. 2016a) to estimate the average greedy matching score l-s between storyline words and generated

[7] We compare the plan-and-write methods with their corresponding baselines and with each other. For fairness, the two stories are pooled and randomly permuted. Five judgments are required to reduce the variance in estimation.

[8] The four aspects are each evaluated.

[9] We applied qualification filters that only allow users who have at least 500 previous jobs and had greater than 98% acceptance rate to participate in our survey.

		Title: Computer
Baselines	Inc-S2S	Tom's computer broke down. He needed to buy a new computer. He decided to buy a new computer. Tom bought a new computer. Tom was able to buy a new computer.
	Cond-LM	The man bought a new computer. He went to the store. He bought a new computer. He bought the computer. He installed the computer.
Dynamic	Storyline	needed → money → computer → bought → happy
	Story	John needed a computer for his birthday. He worked hard to earn money. John was able to buy his computer. He went to the store and bought a computer. John was happy with his new computer.
Static	Storyline	computer → slow → work → day → buy
	Story	I have an old computer. It was very slow. I tried to work on it but it wouldn't work. One day, I decided to buy a new one. I bought a new computer .
		Title: The Virus
Baselines	Inc-S2S	His computer was fixed and he fixed it. John got a new computer on his computer. John was able to fix it himself. John was able to fix his computer and was able to fix his computer. John was able to fix his computer and had a virus and was able to fix his computer.
	Cond-LM	Tim was working on a project. He was working on a project. Tim was working on a project. The project was really good. Tim was able to finish the project.
Dynamic	Storyline	computer → use → anywhere → house → found
	Story	I was working on my computer today. I was trying to use the computer. I couldn't find it anywhere. I looked all over the house for it. Finally, I found it.
Static	Storyline	work → fix → called → found → day
	Story	I had a virus on my computer. I tried to fix it but it wouldn't work. I called the repair company. They came and found the virus. The next day, my computer was fixed.

Table 4: Case studies of generated storylines and stories.

Title / Problem	**Story**
Taxi / off-topic	I got a new car. It was one day. I decided to drive to the airport. I was driving for a long time. I had a great time .
Cut / repetitive	Anna was cutting her nails. She cut her finger and cut her finger. Then she cut her finger. It was bleeding! Anna had to bandage her finger.
Eight glasses/ inconsistent	Joe needed glasses. He went to the store to buy some. He did n't have any money. He found a pair that he liked. He bought them.

Table 5: Example stories that demonstrate the typical problems of the current systems.

Method	l-B1	l-B2	l-s
Dynamic	6.46	0.79	0.88
Static	9.53	1.59	0.89

Table 6: The storyline BLEU score (only BLEU-1 and BLEU-2) and the correlation of storyline-story l-s.

story sentences. Concretely, a storyline word is greedily matched with each token in a story sentence based on the cosine similarity of their word embeddings[10]. The highest cosine score is regarded as the correlation between them.

Table 6 shows the results. We can see that the static schema generates storylines with higher BLEU scores. It also generates stories that have a higher correlation with the storylines (higher l-s score[11]). This indicates that with better storylines (higher BLEU score), it is easier to generate more relevant and coherent stories. This partially explains why the static schema performs better than the dynamic schema.

Case study. We further present two examples in Table 4 to intuitively compare the plan-and-write methods and the baselines[12]. In both examples, the baselines without planning components tend to generate repetitive sentences that do not exhibit much of a story progression. In contrast, the plan-and-write methods can generate storylines that follow a reasonable flow, and thus help generate coherent stories with less repetition. This demonstrates the ability of the plan-and-write methods. In the second example, the storyline generated by the dynamic schema is not very coherent and thus significantly affects story quality. This reflects the importance of storyline planning in our framework.

Error analysis. To better understand the limitation of our best system, we manually reviewed 50 titles and the corresponding generated stories from our static schema to conduct error analysis. The three major problems are: off-topic,

[10]For fairness, We adopt the pre-trained Glove embedding to measure the correlation. http://nlp.stanford.edu/data/glove.840B.300d.zip

[11]There are 75% and 78% storyline words appear in the generated stories in the dynamic and static schema, respectively.

[12]More examples please see our live demo at http://cwc-story.isi.edu/